

ELVIS YEBOAH-DUAKO

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Research Interests

Systems Security, Cloud and Serverless Security, Least-Privilege Access Control, AI Agent Security, Intent-Based Networking, Network Security, Information Flow Control, Data Privacy

Education

Ph.D. in Computer Science

2024 – Present

University of Massachusetts Amherst (Manning CICS), Amherst, MA

- Advisor: [Professor Pubali Datta](#)

M.Eng. in Computer Science and Technology

2021 – 2023

University of Electronic Science and Technology of China (UESTC), Chengdu, China

- Thesis: On Securing Resource-Constrained IoT Networks: A Provably Secure and Decentralized Node Authentication and Key Agreement Protocol (Advisor: [Professor Xia Qi](#))

B.Sc. in Telecommunication Engineering

2016 – 2020

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana

- Project: Design of an Automated Toll Collection and Management System (Advisor: [Dr. Emmanuel A. Affum](#))

Research Experience

Research Assistant

Fall 2024 – Present

Manning CICS, University of Massachusetts Amherst

- Investigate systems security and least-privilege authorization across cloud, serverless, IoT, and AI-agent systems, combining empirical measurement with the design and evaluation of security mechanisms.
- Developed a static-analysis framework for measuring excessive IAM permissions in real-world serverless applications; analyzed 3,363 open-source applications, finding 538,390 granted permissions versus 1,896 required permissions and evaluating automated least-privilege policy reduction.
- Develop AI-assisted Intent-Based Networking (IBN) systems using ONOS, OpenFlow, and Mininet, studying how LLMs compile high-level network intents into executable flow rules and comparing AI-generated policies with classical network compilation.
- Characterize failure modes in AI-generated network policies, including incorrect host/port authorization, missing flow selectors, and constraint violations; develop validation and prompt-based mitigations for safer network policy generation.
- Investigate authorization for AI agents, examining how authentication and authorization mechanisms can accommodate dynamic, tool-using agent behavior in cloud environments.

Graduate Research Assistant

2021 – 2023

Institute for Blockchain & Security, UESTC, China

- Conducted systematic literature reviews on secure blockchain applications in Human Resources Management, synthesizing prior approaches and identifying research gaps.
- Contributed to manuscript development through technical writing, organization of research findings, and preparation of materials for IEEE publication.

Research & Teaching Assistant

2020 – 2021

Department of Telecommunications Engineering, KNUST, Ghana

- Investigated machine-learning-based DDoS detection in SDN/NFV environments, evaluating deep-learning and classical machine-learning approaches for network attack detection.
- Designed and conducted network simulations and experiments to evaluate SDN/NFV architectures and their resilience to network attacks.
- Designed and taught an introductory SDN/NFV course combining networking theory with hands-on laboratory exercises.
- Mentored undergraduate students in Computer Networking, Network Planning, and Digital Systems.

Undergraduate Research Assistant

2019 – 2020

Regional Water Environment & Sanitation Center, KNUST (RWESCK), Ghana

- Investigated IoT-based approaches for real-time water quality monitoring in rural Ghana.
- Designed and prototyped a remote water monitoring and purification system, including a web-based interface for monitoring system data.

Publications

Conference Papers

- [C1] D. Kang, **E. Yeboah-Duako**, S. Thomas and P. Datta. “When Apps Outlive Vendors: Security Implications of IoT Abandonware.” *ACM Conference on Computer and Communications Security (ACM CCS 2026)*, To Appear. [\[Paper\]](#)

Under Review

- [S1] **E. Yeboah-Duako** and P. Datta. “Permissions on the Loose: Measuring Overprivilege in Real-World Serverless Applications.” *Under Review: USENIX Security Symposium 2027*. [\[Paper\]](#)

Industry & Engineering Experience

Software Engineer

2022 – 2024

Orgmento (UK) Ltd., Accra, Ghana

- Led migration of 60,000+ user authentication profiles from a legacy identity system to **Azure AD B2C**.
- Designed and built an identity gateway securing access to a client’s suite of tools and services using **ASP.NET Core, OpenID Connect, and LDAP**.
- Developed backend infrastructure enabling non-technical users to execute Python scripts through a web interface.
- Introduced **GitHub Advanced Security** and **SonarCloud** for automated vulnerability and code-quality analysis.

Technical Skills

Programming: Python, JavaScript/TypeScript, C#, Go, SQL

Networking & Systems: TCP/IP, SDN, OpenFlow, ONOS, Mininet, Linux, network simulation and experimentation

Cloud & Security: AWS, IAM, Serverless Framework, Azure AD B2C, Docker, CodeQL, static analysis, access control

Research & AI: LLMs, AI agents, AI-assisted network policy generation, security policy analysis, empirical measurement, Git/GitHub

Selected Talks

Guest Speaker — Enforcing Least-Privilege Authorization in Serverless Cloud Applications

Spring 2025

Women in Cybersecurity (WiCyS), UMass Amherst Chapter

Awards & Honors

Inclusive Leadership Fellowship	<i>Fall 2025</i>
Jim Gray Scholarship in Computer Science	<i>Spring 2025</i>
Spaulding-Smith Fellowship	<i>Fall 2024</i>
UESTC University Graduate Scholarship Award	<i>Fall 2021</i>
Best Instructor, Huawei ICT Academy Ambassador's Award	<i>Spring 2021</i>
Best Graduating Student, Telecommunication Engineering, KNUST	<i>Fall 2020</i>
Outstanding Prize, Huawei ICT Competition — Sub-Saharan Africa Regional Finals	<i>Summer 2020</i>
First Prize, Huawei ICT Competition — National Finals, Ghana (Network Track)	<i>2019</i>

Teaching & Mentoring

Graduate Teaching Assistant	<i>Fall 2026</i>
<i>Computer Networks, UMass Amherst</i>	
Graduate Teaching Assistant	<i>Fall 2025</i>
<i>Introduction to Computer & Network Security, UMass Amherst</i>	
Graduate Teaching Assistant	<i>Spring 2022</i>
<i>Mobile Computing, UESTC, China</i>	
Course Instructor & Facilitator	<i>Spring 2022</i>
<i>Huawei ICT Academy, KNUST, Ghana</i>	
Undergraduate Teaching Assistant	<i>Fall 2021</i>
<i>Computer Networking, KNUST, Ghana</i>	

Service

Judge, Massachusetts Science & Engineering Fair, Middle School Division	<i>Spring 2026</i>
PhD Peer Mentor, Manning CICS, University of Massachusetts Amherst	<i>2025/2026</i>
Graduate Mentor, IDS STEMPOWER Fellowship Program, UMass Amherst	<i>2025/2026</i>